

MODERN EDUCATION SOCIETY'S

NOWROSJEE WADIA COLLEGE

(An Autonomous College under Savitribai Phule Pune University)

Laboratory course on SADP and Microservices

National Education Policy (NEP) Syllabus

To be implemented from Academic Year 2023-2024

M.Sc.(Computer Science)-Part II

Semester III

2026-27

Co-Ordinator

Dr. Reena Bharathi **Nowrosjee Wadia College,Pune**

Prepared by

Prof. Monika Jadhav **Nowrosjee Wadia College,Pune**

Roll No. :
Name :

SOFTWARE ARCHITECTURE AND DESIGN PATTERNS AND MICROSERVICES
ASSIGNMENT COMPLETION SHEET

Sr.No	Assignment Name	Marks
1	**Creational Design Patterns**	
2	**Structural Design Patterns**	
3	**Behavioral Design Patterns**	
4	**Spring Framework**	
5	**Microservices**	
Total out of 25		
Total out of 10 (Convert 25 into 10)		

Signature of Batch In-charge:

Internal Examiner :

External Examiner :

Date :

Assignment 1: Creational Design Patterns

Pre-Reading

- ✔ Concepts of Design Patterns and Basics of Java
- ✔ Creational Design Patterns –Abstract Factory, Factory Method, Singleton

Lab Assignments

1. Write a Java Program to implement Factory method for Pizza Store with createPizza(), orderPizza(), prepare(), Bake(), cut(), box(). Use this to create a variety of pizza's like NyStyleCheesePizza, ChicagoStyleCheesePizza etc.

2. Write a Java Program to implement Singleton pattern for multithreading.

3. Write a Java Program to implement Abstract factory pattern for various functionalities of the mobile phone, such as taking a photo, recording a video.

Extra Practice Assignment :

1. Write a Java Program to implement Abstract factory pattern to create cars and specifications specific to North America, Europe.

2. Write a Java Program to implement Singleton pattern to create the singleton class LoggerService with a private constructor and a public static method to get the instance.

Signature of the instructor _____ Date _____

Assignment Evaluation

0: Not done	☐	2: Late Complete	☐	4: Complete	☐
1: Incomplete	☐	3: Needs Improvement	☐	5: Well Done	☐

Assignment 2 : Structural Design Patterns

Pre-Reading

- ✔ Concept of Design Patterns and Basics of Java.
- ✔ Structural Design Patterns – Adapter, Decorator, Bridge, Proxy

Lab Assignments

1. Write a Java Program to implement I/O Decorator for converting uppercase letters to lowercase letters.
2. Write a Java Program to implement Adapter pattern for Enumeration iterator.
3. Write a Java Program to implement Bridge design pattern to produce and assemble the two different vehicles.

Extra Practice Assignment :

1. Write a java program to implement Adapter pattern to design Heart Model to Beat Model.
2. Write a java program to implement Proxy Pattern which efficiently manages the loading and displaying of images by introducing a proxy that controls access to the real image object.

Signature of the instructor _____ Date _____

Assignment Evaluation

0: Not done ☐ 2: Late Complete ☐ 4: Complete ☐

1: Incomplete ☐ 3: Needs Improvement ☐ 5: Well Done ☐

Assignment 3: Behavioral Design Patterns

Pre-Reading

- ✔ Concept of Design Patterns and Basics of Java.
- ✔ Behavioral Design Patterns – Observer, Command, Iterator, State

Lab Assignments

1. Write a JAVA Program to implement built-in support (java.util.Observable) Weather station with members temperature, humidity, pressure and methods measurementsChanged(), set Measurement(), getTemperature(), getHumidity(), getPressure().
2. Write a Java Program to implement an undo command to test Ceiling fan.
3. Write a Java Program to implement Iterator Pattern for Designing Menu like Breakfast, Lunch or Dinner Menu.

Extra Practice Assignment :

1. Write a Java Program to implement State Pattern for Gumball Machine. Create an instance variable that holds the current state from there, we just need to handle all actions, behaviors and state transitions that can happen. For actions we need to implement methods to insert a quarter, remove a quarter, turning the crank and display gumball.
2. Write a Java Program to implement command pattern to test Remote Control.

Signature of the instructor ☐ Date ☐

Assignment Evaluation

0: Not done	☐	2: Late Complete	☐	4: Complete	☐
1: Incomplete	☐	3: Needs Improvement	☐	5: Well Done	☐

Assignment 4: Spring Framework

Pre-Reading/ Prerequisites

- ✔ Spring Framework
- ✔ Java and Advanced Java
- ✔ Eclipse IDE, JDBC Module

Lab Assignments

1. Design simple HR Application using Spring Framework.
2. Create a task management API where users can manage tasks, categories, and due dates using Spring Framework.
3. Develop a simple user registration and login module using spring boot. Also validates the user registration fields.
4. Write a java program to implement microservice to accept the details of Employee(Eno, Ename, Designation, DeptName,Salary) from a user and store it into the fat file.

Extra Practice Assignment :

1. Create an ERP system that handles inventory, sales, and financial data using Spring Framework.
2. Design a simple inventory system using Spring Framework.

Signature of the instructor ☐ Date ☐

Assignment Evaluation

0: Not done	☐	2: Late Complete	☐	4: Complete	☐
1: Incomplete	☐	3: Needs Improvement	☐	5: Well Done	☐

Assignment 5: Microservices

Pre-Reading / Prerequisites

- ✔ Microservices
- ✔ Spring Framework
- ✔ Java and Advanced Java
- ✔ Eclipse IDE, JDBC Module

Lab Assignments

1. Develop a module as a microservice to create a student management application using spring boot in java.(Refer assignment 4, Q.3)
2. Create a microservice to display the details of all employees of the specific department. (Refer assignment 4, Q.4)
3. Create HR application as a microservices to send the message to the candidate as “You have been selected for an interview”.(Refer assignment 4, Q.1)

Extra Practice Assignment :

1. Create a spring boot application to set up an API gateway into the Student management Microservices project using the Spring Cloud Gateway library.

Signature of the instructor ☐ Date ☐

Assignment Evaluation

0: Not done	☐	2: Late Complete	☐	4: Complete	☐
1: Incomplete	☐	3: Needs Improvement	☐	5: Well Done	☐